HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F70LB

LIFE INSURANCE MATHEMATICS B

IN-PERSON EXAMINATION

April Examination and Assessment Diet 2024

Duration: 2 hours

Total marks available: 60

Answer all six questions

Attached: No additional attachments
Supplied: Formulae and Tables for Examinations of the Institute and Faculty of Actuaries
Calculator Permitted: Yes

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

I understand that infringing this statement would represent an academic misconduct offence subject to disciplinary action according to the University Regulations and Policy regarding Academic Misconduct, with possible significant consequences for degree progression and final degree outcome.

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Question 1

You are given the following information in respect of two lives (x) and (y):

$$\ddot{a}_{xy} = 8$$
$$\ddot{a}_{x} = 9$$
$$\ddot{a}_{y} = 10$$
$$Force\ of\ interest\ \delta = 5\% \text{ each year}$$
$$\mu_{x+t} = 0.01$$
$$\mu_{y+t} = 0.03$$

Calculate the following assuming that the lives are independent:

(a) $\bar{a}_{x|y}$

(2 marks)

(b) $\bar{A}_{\overline{xy}}$

(2 marks)

(c) $\mu_{x+t\,:\,y+t}$

(1 mark)

[Total 5 marks]

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Question 2

You are given the following data on male deaths through injury at ages 20 to 39 in a standard population, in Country A in 2016 and Country B in 2021.

	Male deaths through injury		
	Standard population		
Age last birthday	(A) Population (millions)	(B) Deaths	(C) Deaths per million population
20-24	10	3,000	300
25-29	10	3,000	300
30-34	20	6,000	300
35-39	20	6,000	300
Total (20 to 39)	60	18,000	

	Male deaths through injury		
	Country A in 2016		
Age last birthday	(A) Population (millions)	(B) Deaths	(C) Deaths per million population
20-24	0.18	45	250
25-29	0.19	58	305
30-34	0.18	62	344
35-39	0.18	54	300
Total (20 to 39)	0.73	219	

	Male deaths through injury		
	Country B in 2021		
Age last birthday	(A) Population (millions)	(B) Deaths	(C) Deaths per million population
20-24	1.23	321	261
25-29	1.30	312	240
30-34	1.38	359	260
35-39	1.56	422	271
Total (20 to 39)	5.47	1,414	

Note: Death rates in column (C) of each table are expressed as rates *per million lives*. It is acceptable to give your answers as death rates per million lives. You do not need to divide through by a million unless you wish.

(a) Calculate the crude death rate for Country A in 2016 and Country B in 2021.

(2 marks)

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(b) Calculate the standardised death rate for Country A in 2016.

(2 marks)

(c) Calculate the standardised mortality ratio for Country B in 2021.

(2 marks)

(d) State with reasons two types of selection which are relevant to the comparison.

(2 marks)

[Total 8 marks]

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Question 3

An insurer issues 2-year unit-linked policies with the following features.

- Sum assured on death: The greater of £3000 and the bid value of the policyholder's units, paid at the end of year of death
- Premiums: £2000 at the start of each year
- Allocation factor:
  0.97 in year 1
  1.02 in year 2
- Bid\offer spread: 4%
- Policy fee: £25 paid at the start of each year
- Fund management charge: 1.5% of the fund at the end of the year (before any mortality charge is deducted), paid at the end of each year
- Mortality charge: 1% of the sum at risk, deducted at the end of each year (after the fund management charge is deducted)

Basis

- Mortality:

t	q_{x+t}
0	0.002
1	0.003

- Unit fund rate of return:
  +1% in the first year
  -3% in the second year
- Sterling fund rate of return: 1% a year
- Initial expenses: £250 at the start of year 1
- Renewal expenses: £20 at the start of year 2
- Fund management expenses: 0.5% of the fund at year end (after deduction of the fund management charge and the mortality charge)

Calculate the profit vector and profit signature for this policy.

[Total 14 marks]

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Question 4

Briefly explain how the risks are different for the member of a defined benefit pension scheme compared to a defined contribution scheme.

[Total 2 marks]

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Question 5

A life office uses a competing risk model with three states labelled 0, 1 and 2. The transition intensities are $\mu_t^{01} = 1$ and $\mu_t^{02} = 2$. All other transition intensities are zero.

(a) State the three assumptions normally used for a Markov multi-state model.

(3 marks)

(b)

(i) Prove from first principles that in this model

$$\frac{d}{dt}\left({}_tp_x^{00}\right) = -3 \ {}_tp_x^{00}$$

(ii) State, with reasons, which assumptions from part (a) were needed in your proof in part (b)(i) and which were not needed.

(6 marks)

(c) State the Kolmogorov forward equations for ${}_tp_x^{01}$ and ${}_tp_x^{02}$.

(1 mark)

(d) Use your answers to (b) and (c) in an Euler scheme with step size $h = 0.1$ to estimate ${}_{0.2}p_x^{01}$.

(3 marks)

(e) Use your answer to (c) to solve ${}_tp_x^{01}$ and hence calculate the exact value of ${}_{0.2}p_x^{01}$.

(3 marks)

(f) Briefly explain why we usually use numerical methods (not exact solutions) to find probabilities in a multi-state model.

(1 mark)

[Total 17 marks]

PLEASE TURN OVER

Question 6

An insurer offers Income Protection Insurance (IPI) policies for terms up to 10 years. It uses a model with four states as follows:

- State A ("Active"): The policyholder has a job and is in good health
- State U ("Unemployed"): The policyholder does not have a job but is in good health
- State S ("Sick"): The policyholder is ill and cannot work
- State D ("Dead"): The policyholder has died.

All policyholders start in state A at time zero. The annual rate of premium is £1,000.

The insurer estimates the following transition intensities:

$$\mu_t^{AU} = 0.2\left(1 - \frac{t}{10}\right)$$

$$\mu_t^{UA} = 0.5$$

$$\mu_t^{AS} = \mu_t^{US} = 0.03$$

$$\mu_t^{SA} = 0.1$$

$$\mu_t^{AD} = \mu_t^{SD} = \mu_t^{UD} = 0.01t$$

All other transition intensities are zero.

(a) Draw a multi-state diagram of this model. Label the states A, U, S and D. Include the transition intensities listed above on your diagram.

(3 marks)

(b) Explain with reasons whether the transition intensities seem suitable for this model and suggest how the model could be improved.

(4 marks)

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(c) Explain in words what each of the following integrals represents. You may assume that positive cashflows are payments from policyholder to the insurer and that negative values are the opposite. (Hint: You may find it useful to use some, but not all, of the terminology “deferred period”, “waiting period”, “off period”, and “waiver of premiums”.)

(i) $1{,}000 \int_0^{10} {}_tp_x^{AA}\, e^{-\delta t} dt$

(1 mark)

(ii) $-100{,}000 \int_0^{10} \left({}_tp_x^{AA}\, \mu_{x+t}^{AD} + {}_tp_x^{AU}\, \mu_{x+t}^{UD} + {}_tp_x^{AS}\, \mu_{x+t}^{SD} \right) e^{-\delta t} dt$

(2 marks)

(iii) $-1{,}000 \int_1^{10} {}_tp_x^{\overline{AA}}\, \mu_{x+t}^{AU}\, e^{-\delta t} \int_0^{10-t} {}_zp_{x+t}^{\overline{UU}} e^{-\delta z} dz\, dt$

(2 marks)

(iv) $-25{,}000 \int_0^{10} \left({}_tp_x^{AA}\, \mu_{x+t}^{AS} + {}_tp_x^{AU}\, \mu_{x+t}^{US} \right) e^{-\delta t} \int_{0.25}^{\infty} {}_zp_{x+t}^{\overline{SS}} e^{-\delta z} dz\, dt$

(2 marks)

[Total 14 marks]

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